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Original Article

Distinct neural activations correlate with maximization of reward magnitude versus frequency

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Choice selection strategies and decision-making are typically investigated using multiple-choice gambling paradigms that require participants to maximize expected value of rewards. However, research shows that performance in such paradigms suffers from individual biases towards the frequency of gains such that users often choose smaller frequent gains over larger rarely occurring gains, also referred to as melioration. To understand the basis of this subjective tradeoff, we used a simple 2-choice reward task paradigm in 186 healthy human adult subjects sampled across the adult lifespan. Cortical source reconstruction of simultaneously recorded electroencephalography suggested distinct neural correlates for maximizing reward magnitude versus frequency. We found that activations in the parahippocampal and entorhinal areas, which are typically linked to memory function, specifically correlated with maximization of reward magnitude. In contrast, maximization of reward frequency was correlated with activations in the lateral orbitofrontal cortices and operculum, typical areas involved in reward processing. These findings reveal distinct neural processes serving reward frequency versus magnitude maximization that can have clinical translational utility to optimize decision-making.

Key words: reward expected value; melioration; memory; entorhinal cortex; orbitofrontal cortex.

Introduction

Cognitive and neural responses to reward and risk are quintessential to understanding human behavior. Gambling tasks predominantly form the experimental test beds for measuring reward and risk processing abilities in humans (Bechara et al. 1997; Brevers et al. 2013). However, it is widely debated how well these tasks can separate decision-making based on frequency of gains/losses versus expected value (EV) or cumulative payoff for different choice-sets (Bechara et al. 1997, 2000; Garon et al. 2006; Roca et al. 2008; Christakou et al. 2009; R. Mueller et al. 2010; Gupta et al. 2011; Oberg et al. 2011; Must et al. 2013; Bembich et al. 2014; Woodrow et al. 2019).

Bechara et al. (1994) and Bechara and Damasio (2005) introduced the Iowa Gambling Task (IGT) as a neuropsychological probe that is able to evaluate cognitive processes responsible for executing advantageous choices when presented with decks containing different proportions of rewards and penalties (Bechara et al. 1994; Bechara and Damasio 2005). However, Soochow gambling studies that have controlled for gain frequency within decks of the Iowa Gambling Task (IGT) show that subject choices reflect their inherent gain frequency preferences in the task, and portray relatively immediate reinforcement based choice behavior in contrast to behavior that maximizes long-term cumulative reward or EV (Chiu and Lin 2007; Lin et al. 2007; Chiu et al. 2008; Lin et al. 2009; Napoli and Fum 2010; Singh and Khan 2012). Some researchers suggest such local minima choices also referred to as melioration may be the rational solution in the face of uncertainty (Sims et al. 2013). In addition, task related performance measures for long-term EV versus gain frequency bias may reflect distinct reinforcement learning and decision-making strategies, e.g. sensitivity to rewards and risks, learning rate, and other behavioral execution strategies (Franken and Muris 2005; Newman et al. 2008; Furl 2010; Weller et al. 2010; Harman 2011), which prior neuro-cognitive studies of EV maximization have not accounted for.

In this study, we uniquely separate immediate gain frequency bias driven decision-making from advantageous longer-term EV-based decision-making. Specifically, we designed a 2-choice paradigm with 2 distinct blocks—a Δ_0EV (baseline) block where 2 reward choice options have equal EVs and reward variance suitable for measuring the gain frequency bias, and a ΔEV (difference) block where the 2 choice options have unequal EVs suitable for measuring cumulative reward magnitude-based choices. We thereby, tease apart measurements of gain frequency biased response from EV-based response, to understand the distinct cognitive and neural mechanisms underlying these decisions.

Previous neuroimaging studies have suggested the significant role of frontal executive regions that perform reward value evaluations and action control for understanding subjective choices (Bechara et al. 1997, 2000; Aron et al. 2007; Furuyashiki et al. 2008; Kennerley et al. 2011; Brevers et al. 2013; Rich and Wallis 2014; Moccia et al. 2017; Balasubramani and Hayden 2018). On the other hand, integration of rewards over time and making advantageous

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decisions with respect to reward magnitudes have been shown to involve working memory correlates (Hasselmo and Stern 2006; Buzsáki and Moser 2013; Schultz et al. 2015; Katshu et al. 2017; Dundon et al. 2018; Steiger et al. 2019). Uniquely, in this study, we simultaneously estimate the neural correlates for both reward relevant parameters i.e. reward magnitude and reward frequency.

Therefore in this paper, we first aim to understand the behavioral choice outcomes of different reward manipulations presented in 2 separate blocks; specifically in the baseline block 2 choice-sets differ in reward gain frequency but not overall reward magnitude over time or expected value (EV), whereas in the experimental block the 2 choice sets differ in gain frequency like the baseline block and also in EV. We investigate whether choice decision-making in these blocks align with known behavioral strategies such as staying with the same choice after receiving a win or reactively shifting after a loss (Bechara et al. 1994; Cassotti et al. 2011; Forder and Dyson 2016). By comparing how behavior differs in the experimental block from the baseline block, we aim to understand the reward EV effects after controlling for reward frequency. Specifically, we hypothesize that win-stay behavior may be stronger in the experimental block for the higher EV condition. Secondly, we aim to understand whether subject demographics, including age, gender, race, ethnicity, and socio-economic status, as well as mental health, including symptom scores of anxiety, depression, inattention and hyperactivity reported in healthy adults, affect behavioral choices. Experimental block presentation order was controlled for in these analyses. We note that these analyses are motivated by prior research, which has suggested differential effects of gender on response selection of EV choices (Meyers-Levy and Maheswaran 1991; Reavis and Overman 2001; Peresie 2004; Orsini and Setlow 2017). Furthermore, age effects have been found showing older adults have compromised prediction abilities for EV (Reavis and Overman 2001; Cassotti et al. 2011). Several studies have also suggested that mental health has significant effects on choice behavior, for instance, individuals with clinical depression show a deficit in integrating rewards over time (Sheline et al. 2009). Similarly, impairments in decision-making are also observed in cases of anxiety and attention deficit hyperactivity disorder (ADHD) (Lawton and Kallai 2002; Herrero et al. 2006; Kalisch et al. 2006; Bremner et al. 2007; Hartley and Phelps 2012; Mäntylä et al. 2012). Finally, we aimed to understand the neural correlates of the behavioral choice-making, specifically when choices are driven by maximization of reward magnitude (EV) versus when driven by reward frequency, and also whether these neural correlates are influenced by the demographic and mental health variables.

Methods

Participants

In total, 198 adult human subjects (age mean ± standard deviation 35.44 ± 20.30 years, range 18–80 years, 115 females) participated in the study. All participants provided written informed consent for the study protocol approved by the University of California San Diego institutional review board (UCSD IRB #180140). Twelve of these participants were excluded from the study as they had a current diagnosis for a psychiatric disorder and current/recent history of psychotropic medications for a final sample of 186 healthy adult participants. All participants reported normal/corrected-to-normal vision and hearing and no participant reported color blindness. For older adults >60 years of age, participants were confirmed to have a Mini-Mental State Examination (MMSE) score > 26 to verify absence of apparent cognitive impairment (Arevalo-Rodriguez et al. 2015). All data were collected prior to the coronavirus disease of 2019 period of restricted human subjects research.

Surveys

All participants provided demographic information by self-report including age, gender, race (in a scale of 1–7: Caucasian; Black/African American; Native Hawaiian/Other Pacific Islander; Asian; American Indian/Alaska Native; More than one race; Unknown or not reported) and ethnicity; socio-economic status (SES) was measured on the Family Affluence Scale from 1 to 9 (Boudreau and Poulin 2008), and any current/past history of clinical diagnoses and medications were reported. For older adults >60 years of age, participants completed the Mini-Mental State Examination (MMSE) and scored >26 to verify absence of apparent cognitive impairment (Arevalo-Rodriguez et al. 2015). All participants completed subjective mental health self-reports using standard instruments, ratings of inattention and hyperactivity obtained on the Adult ADHD Rating Scale (New York University and Massachusetts General Hospital. Adult ADHD-RS-IV with Adult Prompts. 2003; 9–10), Generalized Anxiety Disorder 7-item scale GAD-7 (Spitzer et al. 2006) and depression symptoms reported on the 9-item Patient Health Questionnaire, PHQ-9 (Kroenke et al. 2001). Symptoms for these psychiatric conditions were measured because they have been related to changes in reward processing (Luman et al. 2010; Dillon et al. 2014; Admon and Pizzagalli 2015).

Task design

We investigated a 2-choice decision-making task that enabled a rapid assessment and was easy to understand across the adult lifespan. In this task that we refer to as *Lucky Door*, participants were given the below instruction:

> "You will see two doors.
> Tap left or right to choose a door.
> You will gain or lose coins at each door. Choose the lucky door."

Thus, participants chose between 1 of 2 doors, either a rare gain door (RareG, probability for gains $P = 0.3$, for losses $P = 0.7$) or a rare loss door (RareL, probability for losses $P = 0.3$, for gains $P = 0.7$). Participants used the left and right arrow keys on the keyboard to make their door choice. Door choice was monitored throughout the task. The task choice decisions on each trial were response-constrained, not time-constrained, i.e. participants could take their time to select their choice.

In 2 separate blocks, we investigated whether the overall expected value (EV) of the choice door can influence individual behavior. In the baseline block with Δ_0EV (no-EV difference), the 2 choice doors did not differ in EV (RareL door, $P = 0.3$ for −70 coins and $P = 0.7$ for +30 coins, EV = 0; RareG door, $P = 0.3$ for +70 coins and $P = 0.7$ for −30 coins, EV = 0). In the experimental difference block with ΔEV (EV difference), expected value or EV was greater for the RareG door ($P = 0.3$ for +60 coins, $P = 0.7$ for −20 coins, EV = +40) than for the RareL door ($P = 0.3$ for −60 coins, $P = 0.7$ for +20 coins; EV = −40). Manipulation of EV, with greater expected value tied to the RareG door, allowed for investigating individual propensities to prioritize long-term (or cumulative) versus short-term (or immediate) rewards. The RareG door was assigned greater EV because selecting this door suggests EV magnitude-based decision processing in subjects as opposed to simply choosing based on frequency of gains, in which case the RareL choice should be preferred. Forty trials were presented per block approximating similar trial numbers

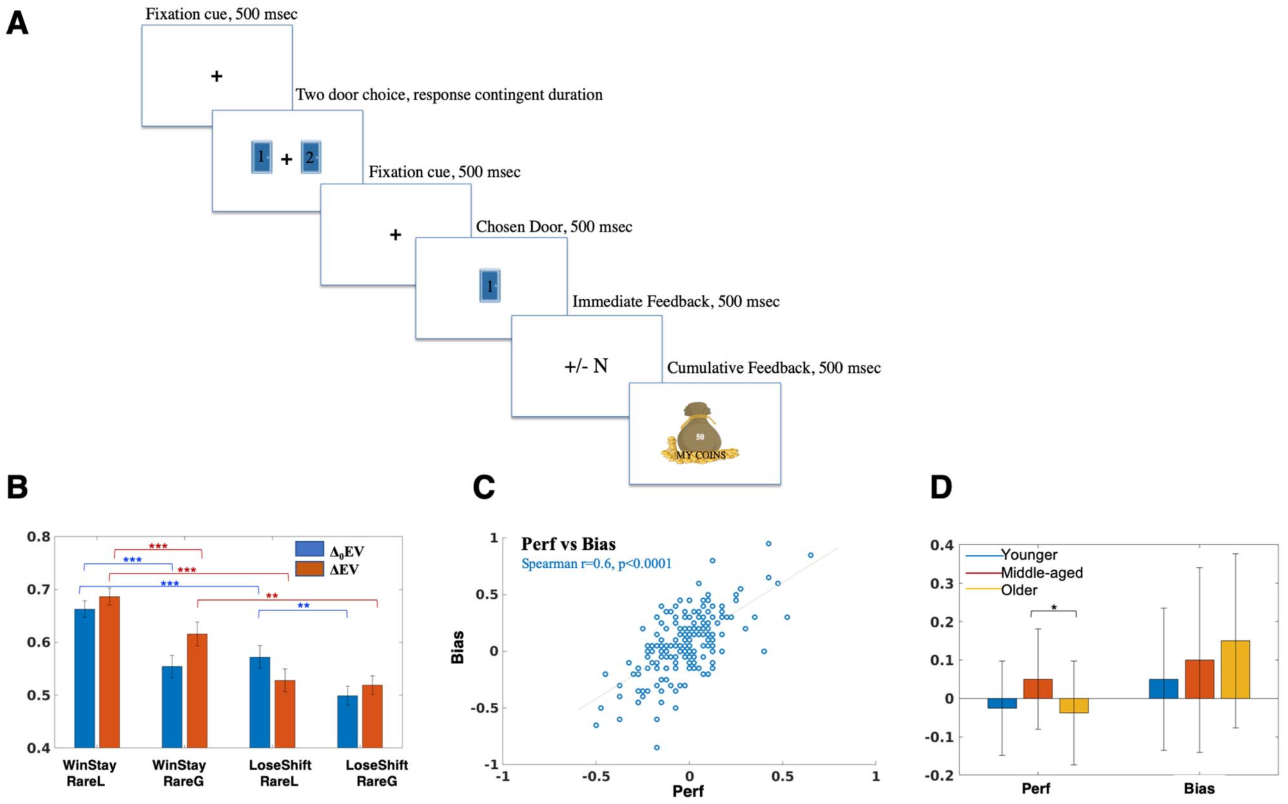

Fig. 1. Reward task and associated behavior. A) Task schematic showing central fixation for 0.5 s followed by 2 choice doors. Post-response, fixation reappeared for 0.5 s, followed by presentation of the chosen door for 0.5 s, then gain or loss trial reward feedback provided for 0.5 s, and finally, cumulative feedback of all gains/losses up to the present trial shown for 0.5 s. Reward distributions for the door choices are presented in Table 1. B) Win-Stay and Lose-Shift difference as seen as a function of the 2 block types and RareL/RareG choices with *:$P < 0.05$, **:$P < 0.01$, ***:$P < 0.005$. RareG choices only in the ΔEV block showed significantly greater Win-Stay versus Lose-Shift behavior. C) Reward EV performance (*Perf*) and reward frequency bias (*Bias*) measures are significantly correlated. D) *Perf* but not *Bias* differ by age categories, specifically middle-aged adults (26–60 years old) had significantly greater EV maximization (*Perf*) than older adults (>60 years old). The bar graphs present median and MAD error bars of the *Perf* and *Bias* variables for the 3 age groups.

as previous human reward task studies (Lin et al. 2009; Bress et al. 2012). Block order was randomized across participants, and 2 practice trials preceded the main ΔEV or Δ_0EV blocks. Figure 1A shows a schematic of the task stimulus sequence. On each task trial, a fixation cue was followed by 2 door choices that remained on the screen until a choice was made. After choice selection, central fixation was presented for 500-ms duration followed by selected choice presentation for 500-ms duration, then trial reward presentation for 500-ms duration corresponding to the reward for the selected door on that trial, and then cumulative reward presentation corresponding to total reward earned until that trial during the block for 500-ms duration. Table 1 shows the reward distribution that was shuffled and updated after every 10 trials had been sampled from that set.

The *Lucky Door* task was deployed in Unity as part of the assessment suite on the *BrainE* (short for Brain Engagement) platform (Balasubramani et al. 2021). The Lab Streaming Layer (LSL; Kothe et al. 2019) protocol was used to time-stamp each stimulus/response event during the task. Study participants engaged with the assessment on a Windows 10 laptop sitting at a comfortable viewing distance.

Table 1. Reward distributions for the door choices in the ΔEV and Δ_0EV blocks.

	ΔEV		Δ_0EV	
	RareL	RareG	RareL	RareG
1	20	−20	30	−30
2	20	−20	30	−30
3	20	−20	30	−30
4	20	−20	30	−30
5	20	−20	30	−30
6	20	−20	30	−30
7	20	−20	30	−30
8	−60	60	−70	70
9	−60	60	−70	70
10	−60	60	−70	70
Sum	−40	40	0	0
Average	−4	4	0	0
Variance	1,493.33	1,493.33	2,333.33	2,333.33

The 2 door choices in either block were RareG (rare gains and frequent losses) and RareL (rare losses and frequent gains). EV (expected value) for RareG and RareL were the same in the Δ_0EV block and greater for RareG relative to RareL in the ΔEV block. RareG and RareL distributions had the same sum, average and variance in the Δ_0EV block, and different sum and averages but same variance in the ΔEV block.

Electroencephalography

Electroencephalography (EEG) data were collected simultaneous to the *Lucky Door* task using a 24-channel Smarting device with a semi-dry and wireless electrode layout (Next EEG—new human interface, MBT). Data were acquired at 500-Hz sampling frequency at 24-bit resolution. Cognitive event markers were integrated using LSL and data files were stored in xdf format.

Behavioral analyses

Choice response speeds on the *Lucky Door* task were calculated as log(1/RT), where RT is response time in seconds. We computed the EV magnitude sensitive performance response (*Perf*) as the difference in proportion selection of the RareG door between the ΔEV and the Δ_0EV blocks; EV of the presented door choices differed only in the ΔEV block, whereas there were no EV differences in the Δ_0EV block (see equation 1 below). For N fraction of responses in each block, we calculated:

$$Perf = \mathrm{N} : \mathrm{RareG}_{\Delta\mathrm{EV}} - \mathrm{N} : \mathrm{RareG}_{\Delta 0\mathrm{EV}} \tag{1}$$

We computed gain frequency bias (*Bias*) as the difference in selected proportion of RareL and RareG doors in the Δ_0EV block where the EV for both the doors was the same. Thus, although *Perf* is indicative of subjective EV maximization, *Bias* informs the choice bias and melioration for higher immediate gain frequencies.

$$Bias = \mathrm{N} : \mathrm{RareL}_{\Delta 0\mathrm{EV}} - \mathrm{N} : \mathrm{RareG}_{\Delta 0\mathrm{EV}} \tag{2}$$

We also calculated Win-Stay and Lose-Shift performance on both ΔEV and Δ_0EV blocks. Win-Stay was computed as the proportion of times the subject repeated the same choice option in the next trial after obtaining a gain for choosing that option in the current trial. Lose-Shift was computed as the proportion of times the subject would shift away from the current choice option in the next trial, on obtaining a loss in the current trial.

Neural data processing

We applied a uniform processing pipeline to all EEG data acquired simultaneous to the reward task. This included: data preprocessing, computing event-related spectral perturbations (ERSP) for all channels, and cortical source localization of the EEG data filtered within relevant theta, alpha and beta frequency bands.

Detail steps include:

(i) EEG data were resampled at 250 Hz, and filtered in the 1–45 Hz range to exclude ultraslow direct current (DC) drifts at <1 Hz and high-frequency noise produced by muscle movements and external electrical sources at >45 Hz.

(ii) EEG data were average referenced and epoched to the chosen door presentation during the task, in the −500 ms to +1,500 ms time window (Fig. 1).

(iii) Any missing channel data (one channel each in 6 participants) was spherically interpolated to nearest neighbors using pop_interp function of MATLAB.

(iv) Epoched data were cleaned using the autorej function in EEGLAB to remove noisy trials (>5sd outliers rejected over max 8 iterations; $0.91 \pm 2.65\%$ of trials rejected per participant). EEG data were further cleaned by excluding signals estimated to be originating from non-brain sources, such as electrooculographic, electromyographic, or unknown sources, using the Sparse Bayesian learning (SBL) algorithm (Ojeda et al. 2018, 2021; https://github.com/aojeda/PEB); this algorithm has the option to output clean EEG data after non-brain source artifacts have been removed. The SBL algorithm is explained further in the cortical source localization step below.

(v) Cortical source localization was performed to map the underlying neural source activations for the ERSPs using the block-Sparse Bayesian learning (SBL) algorithm (Ojeda et al. 2018, 2021) implemented in a recursive fashion. This is a 2-step algorithm in which the first-step is equivalent to low-resolution electromagnetic tomography (LORETA; Pascual-Marqui et al. 1994). LORETA estimates sources subject to smoothness constraints, i.e. nearby sources tend to be co-activated, which may produce source estimates with a high number of false positives that are not biologically plausible. To guard against this, SBL applies sparsity constraints in the second step wherein blocks of irrelevant sources are pruned. Source space activity signals were estimated and then their root mean squares were partitioned into (i) regions of interest (ROIs) based on the standard 68 brain region Desikan–Killiany atlas (Desikan et al. 2006) using the Colin-27 head model (Holmes et al. 1998) and (ii) artifact sources contributing to EEG noise from non-brain sources (such as electrooculographic, electromyographic, or unknown sources); activations from non-brain sources were removed to clean the EEG data. The SBL graphical user interface (GUI) accessible through EEGLAB provides access to an EEG artifact dictionary; this dictionary is composed of artifact scalp projections and was generated based on 6,774 Independent components (IC) available from running Infomax Independent Component Analysis (ICA) on 2 independent open-access studies (http://bnci-horizon-2020.eu/database/data-sets, study id: 005-2015 and 013-2015). The *k*-means method is used to cluster the IC scalp projections into brain, electrooculographic (EOG), electromyographic (EMG), and unknown components. We checked visually that EOG and EMG components had the expected temporal and spectral signatures according to the literature (Jung et al. 2000).

The data-driven sparsity constraint of the SBL method reduces the effective number of sources considered at any given time as a solution, thereby reducing the ill-posed nature of the inverse mapping. In other words, one can either increase the number of channels used to solve the ill-posed inverse problem or impose more aggressive constraints on the solution to converge on the source model. The 2-stage SBL produces evidence-optimized inverse source models at 0.95AUC (area under the curve) relative to the ground truth while without the second stage <0.9AUC is obtained (Ojeda et al. 2018, 2021). Prior research has also shown that sparse source imaging constraints can be soundly applied to low channel density data (Ding and He 2008; Stopczynski et al. 2014). Furthermore in Balasubramani et al. (2021) we have shown that the regions of interest (ROI) estimates resulting from this cortical source mapping have high test–retest reliability (Cronbach's alpha = 0.77, $P < 0.0001$).

(i) For ERSP calculations, we performed time-frequency decomposition of the cleaned epoched EEG data using the continuous wavelet transform (cwt) function in MATLAB's signal processing toolbox. Baseline time-frequency (TF) data in the −250 to −50-ms time window prior to chosen door presentation, which were part of a 500-ms blank fixation period devoid of any cognitive/motor activity, were subtracted from the epoched trials (at each frequency) to observe the event-related synchronization (ERS) and event-related desynchronization (ERD) modulations (Pfurtscheller 1999).

(ii) Cleaned subject-wise trial-averaged EEG data were then specifically filtered in theta (3–7 Hz), alpha (8–12 Hz), and beta (13–30 Hz) bands and separately source localized in each of these bands to estimate their cortical ROI source signals. The source signal envelopes were computed in MATLAB (envelop function) by a spline interpolation over the local maxima separated by at least one time sample; we used this spectral amplitude signal for all neural analyses presented here. We focused on theta, alpha, and beta signals in the relevant selected choice period (0–500 ms after selected choice presentation), trial reward period (500–1,000 ms after selected choice presentation), and the cumulative reward period (1,000–1,500 ms after selected choice presentation). These activations were baseline corrected using signals from the −250 to −50-ms time window corresponding to the blank fixation period prior to chosen door presentation.

Statistical analyses

To investigate behavioral relationships, we fit robust multivariate linear regression models in MATLAB for the *Perf* and *Bias* measures separately (see behavioral analyses for measure descriptions) with the demographic variables (age, gender, race, ethnicity, and SES), order of block presentation, and mental health symptoms as predictors. Robust regression was used to minimize outlier influence (Lane 2002). The dependent response variables in the linear regression models, i.e. *Perf* and *Bias* were log-transformed for normality, and we identified significant factors contributing to the main effects. We report the overall regression model R^2 and P-values, and individual variable β coefficients, t-statistic, degrees of freedom, and P-values.

Channel-wise theta, alpha, beta ERS and ERD modulations for significant spectral activity time-locked to selected choice presentation were computed relative to baseline. Prior to baseline correction, outliers i.e. activations greater than 5MAD from the median across all subjects were removed. The significant average activity across all trials were found by performing t-tests ($P < 0.05$) across subjects, followed by false discovery rate (FDR, alpha = 0.05) corrections applied across the 3 dimensions of time, frequency, and channels (Genovese et al. 2002).

For computing source level activity correlates of the behavioral *Perf* measure, we first found the difference in RareG door specific baseline corrected neural activations between ΔEV and Δ_0EV blocks in 3 frequency bands—theta, alpha, and beta and in 3 relevant trial periods: selected choice presentation, trial reward presentation, and cumulative reward presentation. We used robust linear regression fits for identifying individual ROIs that relate to the *Perf* measure. The independently identified ROIs were further factored in a unified multivariate linear regression model to account for multiple comparisons; significant ROIs in this final multivariate model were reported, $P < 0.05$ (Balasubramani et al. 2021; Grennan et al. 2021). Similarly, for computing source level activity correlates of the behavioral *Bias* measure, we first found the difference in RareL and RareG specific baseline corrected neural activations in Δ_0EV block in 3 frequency bands and in the 3 relevant trial periods: selected choice presentation, trial reward presentation, and cumulative reward presentation. Significant ROIs were identified using robust linear regression fits (fitlm command, robust option in MATLAB) that were further factored in a unified multivariate linear regression model to account for multiple comparisons ($P < 0.05$). For all regression models, standardized beta coefficients were reported to reflect effect sizes, 0.1: small; 0.3: medium; and 0.5 large. Significant interactions were plotted using the plotinteraction function in MATLAB.

Results

Larger EV choices show greater win-stay behavior

Healthy adult subjects ($N = 186$, ages 18–80 years, 115 females) performed a 2-choice gambling task, *Lucky Door*. The task implemented 2 distinct blocks of choices; the Δ_0EV block delivered choice-sets with different gain frequencies but no differences in EV, whereas the ΔEV block delivered choice-sets with same gain frequencies as the Δ_0EV block yet with cumulative long-term reward (EV) differences (Fig. 1A and Table 1). Specifically, the Δ_0EV block only varied the gain frequency associated with the choice doors, with one door leading to 70% positive reward outcomes (Rare Loss or RareL door), whereas the other resulting in 70% negative reward outcomes (Rare Gain or RareG door), yet maintaining the same EV. The ΔEV block, presented in a random sequence order relative to the Δ_0EV block across subjects, had the same gain and loss frequency setup as the Δ_0EV block, but the rewards associated with the RareG door resulted in a larger long-term EV than the RareL door. Participants executed 40 trials per block. Based on performance on the 2 blocks, we calculated an EV related performance measure, *Perf* as the difference in probability of RareG selections (door with positive EV) on ΔEV versus Δ_0EV blocks. Gain frequency *Bias* was measured as the difference in proportion of RareL versus RareG choices on the Δ_0EV block.

In terms of subject behavior, the actual proportion of RareG or RareL choices between the 2 blocks did not differ (paired Wilcoxon signed-rank test: N_{RareG} ΔEV:0.44 ± 0.01; Δ_0EV:0.46 ± 0.01; $P = 0.07$). Overall, all participants chose the higher gain frequency, RareL door more times than the RareG door in both blocks (Δ_0EV block proportion median ± mad RareL: 0.55 ± 0.10, Wilcoxon rank test zval = 4.03, $P = 5.65\text{e}{-5}$; ΔEV block proportion median ± mad RareL: 0.57 ± 0.10, and Wilcoxon rank test zval = 5.47, $P = 4.39\text{e}{-8}$).

Yet, we found that RareG selections (door with long-term EV) were associated with greater Win-Stay behavior on the ΔEV block (Fig. 1B). We set up a 2 × 2 × 2 repeated measures analysis of variance (rm-ANOVA) to understand whether the observed win-stay and lose-shift behaviors differ as a function of block and choice type. The rm-ANOVA with within-subjects factors of block type (ΔEV, Δ_0EV), behavior type (Win-Stay, Lose-Shift), and choice type (RareG, RareL) found significant main effects of behavior type ($F(1, 185) = 27.73$, $P = 3.84\text{e}{-7}$), choice type ($F(1, 185) = 46.75$, $P = 1.14\text{e}{-10}$), but not block type ($P = 0.22$). We also found significant interactions for behavior type x block type ($F(1, 185) = 5.27$, $P = 0.02$), behavior type x choice type ($F(185) = 4.23$, $P = 0.04$), and choice type x block type ($F(1, 185) = 6.61$, $P = 0.01$). Tukey–Kramer post-hoc 3-factor analysis showed significant differences for both Win-Stay ($P = 1.79\text{e}{-6}$) and Lose-Shift ($P = 0.009$) behaviors between RareL (high reward frequency) and RareG (high cumulative reward value) choices in the Δ_0EV block. Win-Stay behavior between RareL and RareG choices in the ΔEV block also differed significantly ($P = 0.0007$). Notably, for RareL choices there was greater Win-Stay than Lost-Shift behavior in both the Δ_0EV block ($P = 0.002$) and ΔEV block ($P = 1.40\text{e}{-8}$). Yet, for RareG choices, there was greater Win-Stay than Lose-Shift behavior only in the ΔEV block ($P = 0.005$).

We then calculated reward EV maximization: *Perf* and gain frequency maximization: *Bias* measures (see Equations (1) and (2) in Methods). The *Perf* and *Bias* variables were significantly

Table 2. Subject characteristics.

Demographics	Median ± MAD
Age	25.00 ± 14.87
Gender *n* (%)	71 (38.17)
Male	115 (61.83)
Female	
Ethnicity *n* (%)	
Caucasian	116 (62.37)
Black or African American	4 (2.15)
Native Hawaiian, Pacific Islander	0 (0)
Asian	37 (19.89)
American Indian, Alaska Native	4 (2.15)
Multi-racial	12 (6.45)
Others	12 (6.45)
Race *n* (%)	
Hispanic or Latino	25 (13.51)
Not Hispanic or Latino	155 (83.78)
Unknown	5 (2.70)
SES	5.00 ± 1.34
Mental Health	Median ± MAD
Anxiety	3 ± 2.88
Depression	3 ± 2.76
Inattention	4 ± 3.99
Hyperactivity	3 ± 2.96
Behavior	Median ± MAD
Perf	−0.02 ± 0.13
Bias	0.10 ± 0.21

Median ± MAD for subjects demographics, mental health self-reports and task behavior variables. MAD, median absolute deviation; SES, socioeconomic status score.

different from each other (Table 2, $P = 2.89e{-}9$) yet were significantly correlated (Spearman $r(185) = 0.60$, $P = 1.53e{-}19$, Fig. 1C).

Choice behavior differs by age categories

We first implemented multivariate linear regression models to check if *Perf* and *Bias* measures had any demographic (age, gender, race, ethnicity, and socio-economic status SES) or mental health (anxiety, depression, inattention, and hyperactivity) predictors in our healthy adult sample (summary scores in Table 2). Order of block presentation was included as covariate in the models. Neither *Perf/Bias* model was significant ($P > 0.1$). Given that age effects have been observed in previous research, we also implemented linear mixed models for *Perf* and *Bias* to understand the fixed effect of age; but this model also did not find a significant effect of age.

Next, we stratified age by 3 different categories of younger adults (18–25 years, $n = 97$); middle-aged (26–60 years, $n = 47$); and older adults (above 60 years, $n = 42$). This stratification is based on the literature on neurodevelopmental trajectories (Lebel et al. 2008; Insel 2014; BrainSpan: Atlas of the Developing Human Brain 2020) and also implemented in our recent research (Grennan et al. 2021). Indeed choice decision response times were different across these age categories, slowest in older adults in both blocks (Δ_0EV block: median ± mad response times: younger 370 ± 88 ms, middle-aged 433 ± 251 ms, older 776 ± 312 ms, Kruskal–Wallis (KW, $df = 183$) chi-sq = 72.8, $P = 1.55e{-}16$; ΔEV block: median ± mad response times: younger 378 ± 129 ms, middle-aged 465 ± 212 ms, older 742 ± 243 ms, KW(183) chi-sq = 64.26, $P = 1.11e{-}14$; posthoc rank tests showed significant between-group response time differences for each age group comparison within each block, $P < 0.02$).

For these 3 age categories, we also found a significant difference in preference for the dominant RareL choices only in the ΔEV block (Δ_0EV block: proportion median ± mad RareL: younger 0.52 ± 0.09, middle-aged 0.55 ± 0.12, older 0.57 ± 0.11, Kruskal–Wallis (183) chi-sq = 3.72, $P = 0.15$; ΔEV block: proportion median ± mad RareL: younger 0.57 ± 0.10, middle-aged 0.52 ± 0.10, older 0.6 ± 0.10, KW(183) = 9.86, $P = 0.0072$). Post-hoc rank tests showed a significant difference only between middle-aged and older groups for RareL preference in the ΔEV block ($P = 0.0054$). Consequently, relative choice proportion comparisons for *Perf* (EV maximization) differed by age categories (median ± mad *Perf*: younger −0.02 ± 0.12, middle-aged 0.05 ± 0.13, older −0.04 ± 0.13, KW(183) = 6.9, $P = 0.03$); with post-hoc rank tests only showing a significant difference between middle-aged and older adults ($P = 0.02$). Whereas *Bias* (gain frequency maximization) did not differ by age categories (median ± mad *Bias*: younger 0.05 ± 0.18, middle-aged 0.10 ± 0.24, older 0.15 ± 0.23, KW(183) = 3.7, $P = 0.16$).

Neural correlates of reward EV magnitude maximization

All participants performed the reward task with simultaneous EEG, which we analyzed in the theta (3–7 Hz), alpha (8–12 Hz), and beta (13–30 Hz) frequency bands. Since frequency band processes can provide greater insight on oscillatory neural mechanisms of cognitive control (Benchenane et al. 2011; Klimesch 2012; Mishra et al. 2012; Cavanagh and Frank 2014; Herrmann et al. 2016), we pursued event-related analyses in the spectral domain. Figure 2A and B shows the trial-baseline corrected event-related spectral perturbations (ERSPs) on the ΔEV versus Δ_0EV blocks for RareG trials that correspond to the EV maximization *Perf* measure. We note that taking the relative block difference in all subjects allows non-task related individual EEG differences to cancel out. Relative responses to the RareG door were important for analysis because this door choice resulted in a larger EV than the other (RareL) door in the ΔEV block, and was also associated with greater Win-Stay than Lose-Shift behavior only in the ΔEV block (above section). In addition, we note that all neural analyses were conducted in time windows post-choice selection so that results are not influenced by age-related response times for choice-making. Specifically, we investigated *Perf* neural activations from 3 relevant trial periods: immediately post-presentation of selected choice but prior to reward (0–500 ms selected choice period), during presentation of immediate trial reward (500–1,000-ms reward period), and during presentation of the cumulative reward up to that trial in the block (1,000–1,500 ms cumulative reward period).

We also implemented cortical source reconstructions of the frequency band data and parcellated cortical source activity as per the Desikan–Killiany regions of interest (Ojeda et al. 2018, 2021; Balasubramani et al. 2021). The corresponding source maps for RareG trials on the ΔEV minus Δ_0EV blocks are shown in Fig. 2C, source maps are corrected for $P < 0.05$ difference between blocks.

To understand which of the observed neural activations were not just significant but also behaviorally relevant, we first generated independent robust linear regression models relating the cortical source activations on RareG trials (ΔEV minus Δ_0EV, observed in Fig. 2C) to *Perf*. To account for multiple comparisons, all independently significant cortical sources were included within a robust multivariate linear regression model predicting *Perf*. This overall multivariate model for *Perf* was significant ($R^2 = 0.30$, $df = 138$, $P = 5.37e{-}8$). The results of this model showed alpha activity in the left parahippocampal cortex during the choice presentation period ($\beta = 0.17 \pm 0.08$, tstat = 2.06, $P = 0.04$) and beta activity in the right entorhinal cortex during the cumulative reward period ($\beta = -0.38 \pm 0.07$, tstat = −5.58, $P = 1.19e{-}7$) as

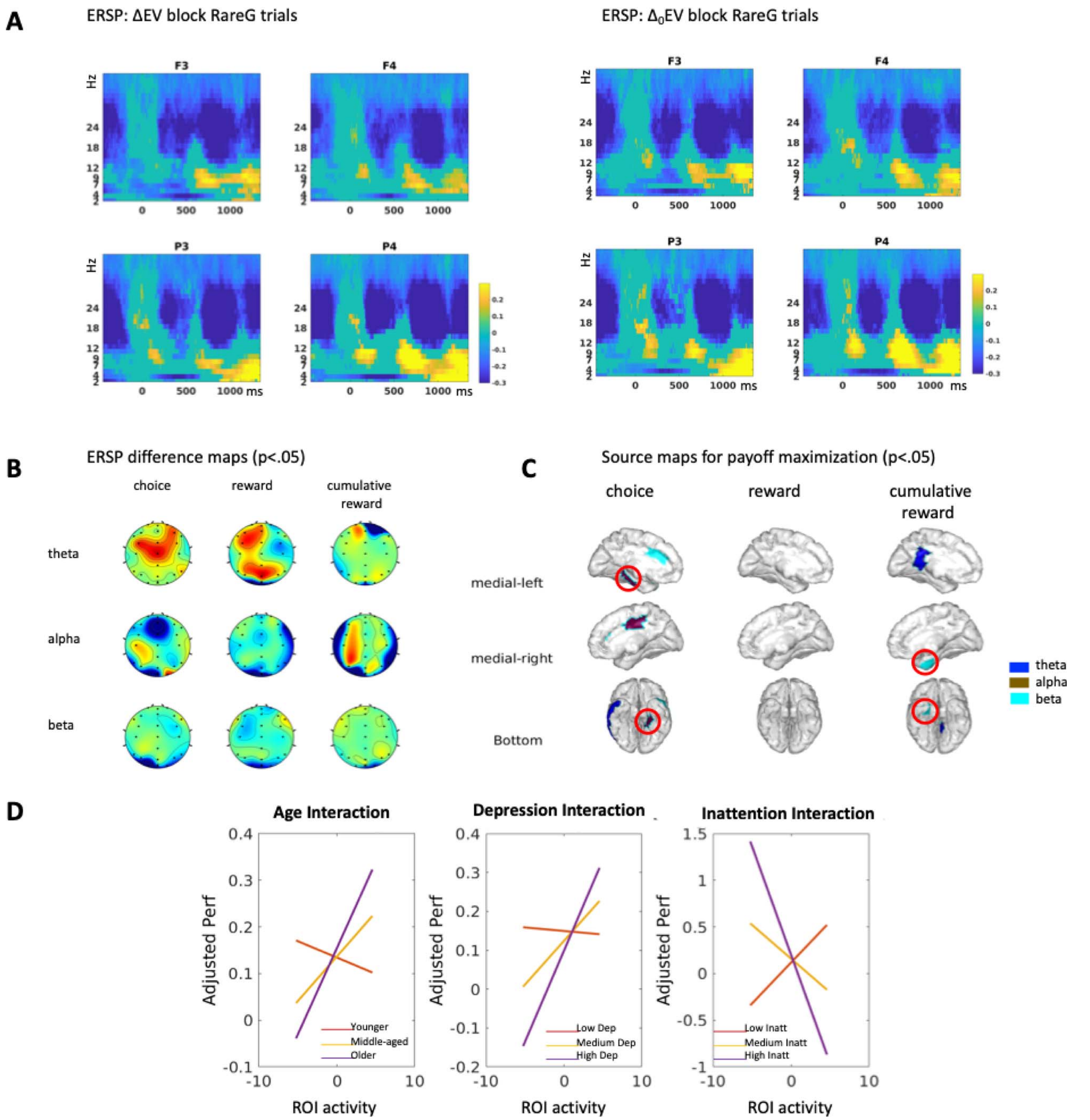

Fig. 2. Neural correlates of reward payoff magnitude maximization. A) shows the ERSPs for RareG trials on the ΔEV and Δ_0EV blocks at frontal (F3, F4) and parietal (P3, P4) electrodes, $P < 0.05$ corrected. ERSPs are baseline corrected and time-locked to choice door presentation at 0–500 ms; immediate reward presented at 500–1,000 ms, and cumulative reward presented at 1,000–1,500 ms. Event-related synchronizations (ERS) is observed in the lower theta/alpha frequency bands and event-related de-synchronizations (ERD) is observed in the beta band. B) shows the ΔEV minus Δ_0EV ERSP difference topography maps, $P < 0.05$ corrected. C) shows the ΔEV minus Δ_0EV source difference maps, $P < 0.05$ corrected. The significant predictors of *Perf* performance: alpha activity in the left parahippocampal cortex during the choice presentation period and beta activity in the right entorhinal cortices during the cumulative reward period are encircled. D) Significant neurobehavioral interactions in the Perf model are shown for beta activity in the right entorhinal cortex during the cumulative reward period; activity in this ROI interacted with age, depression, and inattention.

the most significant predictors of *Perf*, i.e. EV magnitude-based decisions (circled in Fig. 2C).

We further investigated whether these *Perf* neural predictors (alpha activity in the left parahippocampal cortex during the choice presentation period, and beta activity in the right entorhinal cortex during the cumulative reward period) were sensitive to demographic and mental health factors. For this, *Perf* behavior was modeled with predictor of neural activity and neural activity interactions with age categories (younger, middle-aged, and older adults), gender, SES, and mental health self-reports of anxiety, depression, inattention, and hyperactivity, separately for each neural predictor. This overall *Perf* model with neural interactions was significant only for the entorhinal activity during the cumulative reward period ($R^2 = 0.53$, $df = 139$, $P = 3.57e{-}12$); significant interactions are shown in Fig. 2D. We found that entorhinal beta activity interacted significantly with the older age category ($\beta = -1.65 \pm 0.65$, tstat = −2.51, $P = 0.01$) with increase in neural activity in older adults showing a steeper increase in *Perf* relative to other age groups. Entorhinal activity also interacted with depression ($\beta = -1.79 \pm 0.28$, tstat = −6.42, $P = 1.95\text{e}{-9}$); individuals with greater depression scores showed a steep increase in *Perf* with increasing beta activity, whereas individuals with lower depression scores did not show this effect. Finally, entorhinal activity interacted with inattention ($\beta = 1.43 \pm 0.27$, tstat = 5.20, $P = 6.87\text{e}{-7}$); individuals with greater inattention scores showed steeper decrease in *Perf* with increasing beta activity, whereas individuals with lower inattention scores showed the opposite effect.

Neural correlates of reward gain frequency maximization

Calculation of neural activity underlying gain frequency *Bias* corresponded to the behavioral calculation as the difference in responses to the RareL versus RareG choice selections in the Δ_0EV block, which had no expected value difference between the choice doors. Figure 3A shows the trial-baseline corrected ERSPs for RareL versus RareG trials on the Δ_0EV blocks for the *Bias* measure. The corresponding scalp and source difference maps are shown in Fig. 3B and C, corrected for $P < 0.05$ difference between choice doors in the baseline block. As for *Perf*, neural data corresponding to *Bias* was also analyzed in 3 relevant frequency bands, theta, alpha, and beta, and in 3 relevant time windows: immediately post-presentation of selected choice but prior to reward (0–500 ms selected choice period), during presentation of immediate trial reward (500–1,000-ms reward period), and during presentation of the cumulative reward up to that trial in the block (1,000–1,500 ms cumulative reward period). Taking the RareL versus RareG trial contrast allowed for non-task related individual EEG differences to cancel out, and the time windows for neural analyses were all post-choice selection so that results are not influenced by age-related response times for choice-making.

To find behaviorally relevant activations, we first generated independent robust linear regression models relating the significant cortical source activations on RareL minus RareG trials (Δ_0EV block) to *Bias*. Then we accounted for multiple comparisons using a robust multivariate linear regression model that included all independently significant ROI activations. The overall multivariate model for *Bias* was significant ($R^2 = 0.21$, $df = 141$, $P = 0.0004$). In this model, we found that the beta activations in the left lateral orbitofrontal cortex ($\beta = 0.18 \pm 0.07$, tstat = 2.45, $P = 0.01$) and right pars opercularis ($\beta = 0.18 \pm 0.08$, tstat = 2.26, $P = 0.02$), both during the choice presentation period, were significantly related to *Bias* (circled in Fig. 3C). In addition, similar to *Perf*, we checked whether the distinct Bias correlates showed any interactions with age, gender, SES, and mental health self-reports of anxiety, depression, inattention, and hyperactivity, but no significant interactions were found (all $P > 0.05$).

Discussion

Trial-based reinforcement learning models that have immediate reward presented on every trial, suggest that healthy human choices ideally tend to maximize long-term beneficial outcomes (Gupta et al. 2013; Balasubramani et al. 2014, 2015, 2018; Balasubramani and Chakravarthy 2019). However, many existing neuropsychological measures of decision-making that optimize for long-term expected value (EV) may not reliably estimate the participant's ability to integrate rewards and make foresighted decisions, given biased predispositions for frequency of gains and losses that lead to choice of immediate sub-optimal rewards (Stocco et al. 2009; Furl 2010; Napoli and Fum 2010; Gansler et al. 2011). In our paradigm, we study gain frequency driven bias in the Δ_0EV block that has no EV difference between choice options such that decisions are purely based on the frequency of immediate positive rewards (Lin et al. 2009). In contrast, the ΔEV block is designed to have similar reward distribution structure as the Δ_0EV block but differing only on the long-term expected value between choices. Therefore, our study design is uniquely able to distinguish long-term EV magnitude maximization from gain frequency maximization.

We found that the *Perf* measure corresponding to EV maximization and the *Bias* measure corresponding to gain frequency maximization were significantly correlated across subjects. RareL (rare loss) high gain frequency choices were the dominant choices in both blocks, and win-stay behavior was greater than lose-shift for these choices in both blocks. Yet, notably, win-stay behavior was only significantly greater than lose-shift for RareG (rare gain) high EV choices in the ΔEV block, providing a behavioral marker for choice preference for greater EV. We further found that choice-making response times slowed with age, and middle-aged versus older adults had significantly greater *Perf*, i.e. greater RareG preference in the ΔEV versus Δ_0EV blocks. Notably, these results were only found when subjects were divided into 3 age categories, younger (<26 years), middle-aged (26–60 years), and older adults (>60 years); but not when age was treated as a continuous variable. The results showing age category specific differences align with the literature suggesting that older adults have compromised EV prediction (Reavis and Overman 2001; Cassotti et al. 2011).

We then investigated the neural correlates of EV maximization performance (*Perf*) and gain frequency *Bias* using EEG source imaging. Various neuroimaging studies suggest that dorsolateral prefrontal cortex (PFC), ventromedial PFC, and anterior cingulate cortex play key roles in IGT choice behavior. Research also shows that medial prefrontal cortices (mPFC) are activated when subjects experience larger loss magnitudes, and inferior parietal lobule is activated during reward presentation (Lin et al. 2008, 2015). Event-related potential (ERP) recording studies have further identified neural markers underlying an adapted IGT paradigm for choice evaluation, response and reward feedback (Cui et al. 2013); larger amplitude of the ERP P3 component was observed in the left hemisphere when choosing advantageous choices, whereas the P3 was right localized for disadvantageous choices. This study also observed the feedback related negativity (FRN) signal specifically as a function of valence (gain/loss) but not reward magnitude. Another study has suggested a negative polarity ERP sourced from either mPFC or anteriorcingulate cortex (ACC) regions during observation of losses (Gehring and Willoughby 2002). Our paradigm differed from prior ERP studies in that we deployed 2 blocks that either vary in EV between presented choice-sets (ΔEV block) or do not vary in EV (Δ_0EV block). These blocks were presented to understand both EV-related performance, *Perf*, and gain frequency *Bias* preferences. We studied event-related frequency band processes as they can provide greater insight on oscillatory neural mechanisms of cognitive control (Benchenane et al. 2011; Klimesch 2012; Mishra et al. 2012; Cavanagh and Frank 2014; Herrmann et al. 2016).

Our neural results showed that the event-related spectral perturbations last about the entire 500-ms duration when time locked to the selected choice, immediate reward and cumulative reward events, yet the dynamics are distinct for these periods (Figs 2A and 3A). Earlier studies have also suggested complex dynamics mediated by dopamine and serotonin neuromodulation

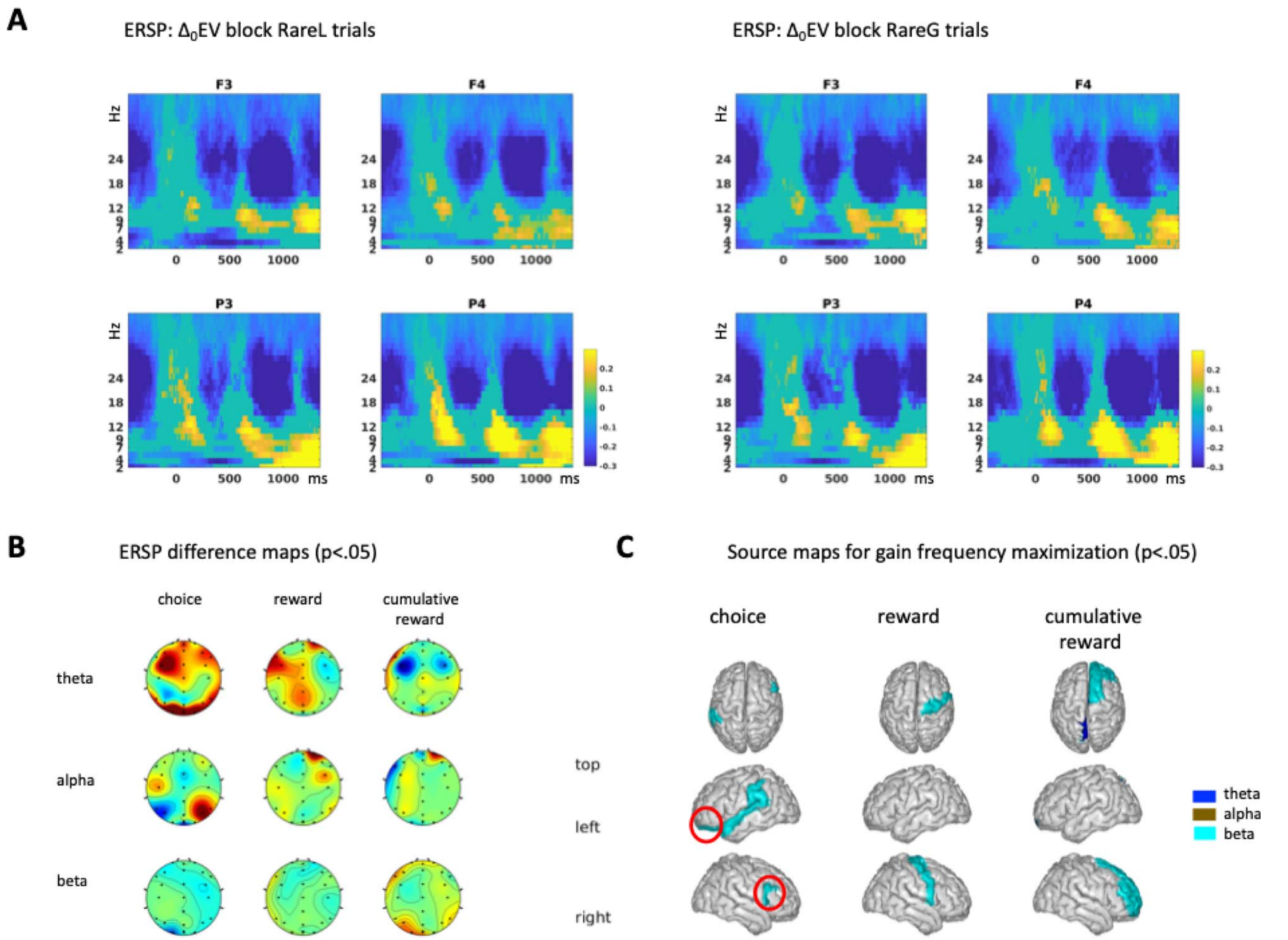

Fig. 3. Neural correlates of reward gain frequency maximization. A) shows the ERSPs for RareL versus RareG trials on the Δ_0EV block corresponding to the *Bias* measure at frontal (F3, F4) and parietal (P3, P4) electrodes, $P < 0.05$ corrected. ERSPs are baseline corrected and time-locked to choice door presentation at 0–500 ms; immediate reward presented at 500–1,000 ms, and cumulative reward presented at 1,000–1,500 ms. B) shows the RareL minus RareG ERSP difference topography maps on the Δ_0EV block, $P < 0.05$ corrected. C) shows the RareL minus RareG source maps, $P < 0.05$ corrected. The significant predictors of reward gain frequency *Bias*: beta activations in the left lateral orbitofrontal cortex and right pars opercularis during the choice presentation period are encircled.

in the cortico-basal ganglia circuit to underlie reward based decision dynamics (Sutton and Barto 1998; Schultz 2000; Doya 2002; Tobler et al. 2009; Balasubramani et al. 2014, 2015, 2018; Chakravarthy et al. 2018). In our study, we did not analyze the actual decision period since 2 different choice options are shown on the screen during this time and the neural signal associated with every choice option was difficult to explicitly assess. All neural analyses were performed post-response and compared 2 different block conditions (RareG trials on ΔEV vs. Δ_0EV block for *Perf*, and RareL vs. RareG trials on Δ_0EV block for *Bias*) so as not to be influenced by age-related response slowing. We further used robust regressions so that significant neural effects could be meaningfully related to behavior (Balasubramani et al. 2021; Grennan et al. 2021, 2022; Shah et al. 2021; Kato et al. 2022). In our analyses, we found distinct neural correlates for *Perf* (EV maximization) and *Bias* (gain frequency maximization).

We found that reward magnitude maximization was most significantly related to activity in memory processing regions, specifically, alpha activity in the left parahippocampal cortex during the choice presentation period corresponding to greater event-related synchronization (ERS) on the RareG trials in the ΔEV versus Δ_0EV block, and beta activity in the right entorhinal cortices during the cumulative reward period corresponding to greater event-related desynchronization (ERD) on the RareG trials in the ΔEV versus Δ_0EV block. This is in line with other studies suggesting the significant role of parahippocampal and entorhinal memory processes in driving specific associative reward integration through binding of relevant task-features and selection of advantageous choices (Gilbert and Fiez 2004; Hasselmo and Stern 2006; Dundon et al. 2018; Steiger et al. 2019; Peng and Burwell 2021). We then asked whether the *Perf* correlates show any interactions with categorical age, gender, other demographics and mental health factors, and found several interactions. The entorhinal activity predictor of EV maximization significantly interacted with older adults age group with greater neural activity showing a steeper increase in *Perf* relative to other age groups. Entorhinal memory processing has been previously shown to be sensitive to interactions with age (Rosen et al. 2003; Jessen et al. 2006). We also found that the entorhinal activity predictor of EV maximization interacted with depression and inattention. Decreased responsiveness to reward is well known in depression (Henriques et al. 1994; Henriques and Davidson 2000; Beevers et al. 2013) and we have also shown reward sensitivity in a recent personalized mood prediction study (Shah et al. 2021). This current study suggests that the entorhinal neural activity predictor of reward magnitude maximization is also sensitive to depressed mood. Our results are also consistent with earlier studies that have demonstrated a role for entorhinal cortices in depression and more generally, emotional episodic memory control (Leal et al. 2017; Lu et al. 2022). Finally, comprehensive meta-analyses in both children and adults with ADHD have shown deficits in reward choice decision-making (Mowinckel et al. 2015; Patros et al. 2016) that may be relevant to the interactions with inattentiveness affecting reward maximization in our healthy sample.

When investigating gain frequency maximization or *Bias*, we observed very different neural correlates from EV maximization. Neural correlates were observed as beta activations in the left lateral orbitofrontal cortex and right pars opercularis during the choice presentation period; these correspond to greater activity synchronization or lesser beta band desynchronization (ERD) on the RareL versus RareG trials in the Δ_0EV block. This evidence aligns with studies suggesting the important role of core frontal executive regions in reward processing and representation of information about corresponding valence (O'Doherty et al. 2001; Aron et al. 2007; Furuyashiki et al. 2008; Rich and Wallis 2014; Du et al. 2020). No neural interactions with demographics or mental health were found for the *Bias* measure.

We also note that we did not observe any significant neural activity related to *Perf* or *Bias* during the immediate reward presentation period. We speculate this may be due to 2 reasons: (i) in the ΔEV block, subjects integrate rewards over time and may find cumulative reward presentation more useful for advantageous decision-making; this is when we find entorhinal beta activity correlates with EV maximization. (ii) It may be that the reward-related information is backpropagated and subjects are able to perform reward prediction during the selected choice presentation period itself rather than the immediate reward period (Schultz et al. 1997), as we observe significant reward-related activity in this period both in the blocks.

Translational neuroscience studies show that reward based decision processing deficits are found in depression and in attention disorders, leading to difficulty in reward integration and foresighted choice-behaviors (Miller and Seligman 1973; Gradin et al. 2011; Groen et al. 2013; Silvetti et al. 2013; Balasubramani et al. 2015; Ziegler et al. 2016; Balasubramani and Chakravarthy 2019). Individuals suffering from such disorders may focus on the reward outcome in the short-term, characterized by a prolonged attenuation of temporal discounting of rewards (Pizzagalli et al. 2005; Eshel and Roiser 2010). Interestingly, our study suggests distinct neural correlates for maximization of the magnitude of reward outcomes versus bias towards repeated selection of frequent gains. We also note that our neurobehavior results, specifically the midfrontal theta activations underlying both EV and bias maximization, and orbitofrontal beta activity underlying gain frequency bias, may be aligned with findings from studies employing reversal learning paradigms (Rolls 1996; Fellows and Farah 2003; Schutte et al. 2017; Rolls et al. 2020). In such paradigms, reward and punishment contingencies associated with choices are varied through time such that choice distributions sometimes have no EV differences, whereas other times EV differences exist akin to our 2 blocks.

Although we study a large adult sample using scalable EEG measures, our results are limited by the trial-averaged EEG analysis in healthy adults, and approximately localized sources for identifying the underlying neural correlates. In future studies, these results need to be replicated in clinical populations and with greater channel density electrophysiology recordings, whereas further exploring single-trial analyses or individual-specific peak frequency analyses. Overall, our study shows that even while EV maximization driven decision-making behavior is correlated with bias towards frequent gains, these processes have distinctly different underlying neural mechanisms. Ultimately, these specific neural processes may have clinical translational application in neuropsychiatric disorders of reward processing.

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Data availability statement

The data will be shared on specific request to the authors.

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